08 Gas phase concentrations

The concentrations of ideal gases can be calculated using the ideal gas equation

$C = \frac{N}{V} = \frac{P}{RT}$

molar flowrate

For a flow reactor, $C = \frac{F}{\upsilon} = \frac{P}{RT}$

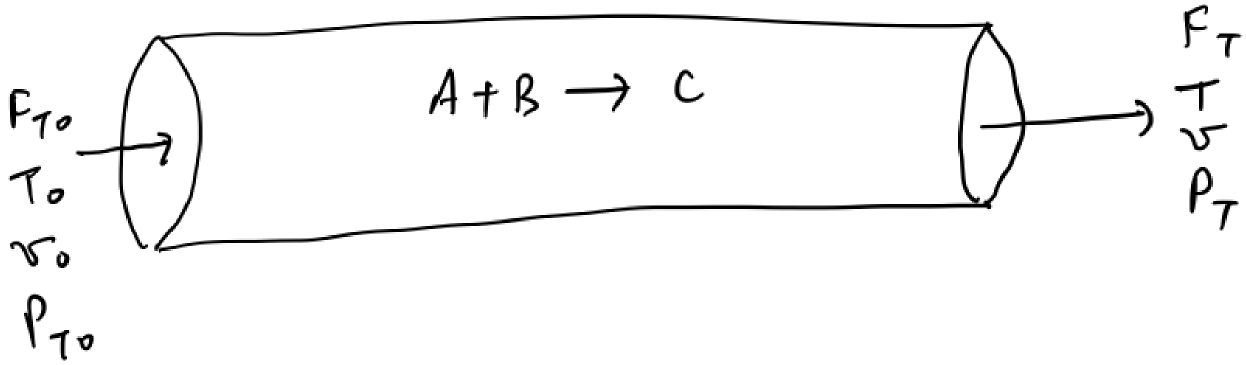
volumetric flowrate

At the entrance to a flow reactor, $C_{T0} = \frac{F_{T0}}{\upsilon_0} = \frac{P_{T0}}{RT_0}$

Where 'T' stands for total

For an ideal gas, $C_T = \sum C_j$ where j is overall species, since
$P_T = \sum P_j$ where P_j are partial pressures

Consider a PFR carrying out the reaction A + B --> C:



This PFR is non-isothermal and experiences pressure drop. How do we compute C_A, C_B, and C_C as functions of conversion?

$$C_j = \frac{F_j}{\upsilon}$$

$$F_j = F_{j0} - \frac{\nu_j}{\nu_{LR}} F_{LR,0} X$$

To get υ, we use "PVT" laws:

$\upsilon = \frac{F_T RT}{P}$

Hence, $\frac{\upsilon}{\upsilon_0} = \dfrac{\frac{F_T R T}{P}}{\frac{F_{T0} R T_0}{P_0}}$ re-arrange

Hence, at any point in the PFR, $\upsilon = \upsilon_0 \left(\frac{F_T}{F_{T0}}\right)\left(\frac{P_0}{P}\right)\left(\frac{T}{T_0}\right)$

$C_j = \frac{F_j}{\upsilon} = \dfrac{F_j}{\upsilon_0 \left(\frac{F_T}{F_{T0}}\right)\left(\frac{P_0}{P}\right)\left(\frac{T}{T_0}\right)}$

Hence, $C_j = \left(\frac{F_{T0}}{\upsilon_0}\right)\left(\frac{F_j}{F_T}\right)\left(\frac{P}{P_0}\right)\left(\frac{T_0}{T}\right) = C_{T0} \left(\frac{F_j}{F_T}\right)\left(\frac{P}{P_0}\right)\left(\frac{T_0}{T}\right)$

In terms of conversion:

Define $\delta = \frac{\text{change in the total number of moles}}{\text{mole of LR reacted}}$

For example, for the reaction aA + bB --> cC + dD, where species A is the limiting reactant: $\delta = \frac{c+d-a-b}{a}$

For example, for the NO oxidation reaction: NO + 1/2 O_2 --> NO_2, $\delta = \frac{1-1-\frac{1}{2}}{1} = -\frac{1}{2}$ (NO is LR)

$\frac{1-1-\frac{1}{2}}{\frac{1}{2}} = -1$ (O_2 is LR)

Then, $F_T = F_{T0} + F_{LR,0}\delta X$ (extent of reaction)

Then $\frac{F_T}{F_{T0}} = \frac{F_{T0} + F_{LR,0}\delta X}{F_{T0}} = 1 + \underbrace{y_{LR,0}\delta}_{\varepsilon} X$

Define $\epsilon = y_{LR,0}\delta$, where $y_{LR,0}$ is inlet mole fraction of the limiting reactant

Then, $\upsilon = \upsilon_0\left(\frac{F_T}{F_{T0}}\right)\left(\frac{P_0}{P}\right)\left(\frac{T}{T_0}\right) = \upsilon_0(1 + \varepsilon X)\left(\frac{P_0}{P}\right)\left(\frac{T}{T_0}\right)$

stoichiometric coefficients - negative for reactants, positive for products

We can do some algebra to arrive at $C_j = \frac{C_{LR,0}\left(\theta_j - \frac{\nu_j}{\nu_{LR}}X\right)}{1+\epsilon X}\left(\frac{P}{P_0}\right)\left(\frac{T_0}{T}\right)$

to do this we sub in $F_j = F_{j0} - \frac{\nu_j}{\nu_{LR}} F_{LR,0} X = F_{LR,0}\left(\theta_j - \frac{\nu_j}{\nu_{LR}} X\right)$

where we have employed $\theta_j = \frac{F_{j0}}{F_{LR,0}}$

Important notes:

① if there is no pressure drop, $\frac{P}{P_0} = 1$.
we do not consider pressure drop in this class, so in this class, P/P_0 always $= 1$.

② $\frac{T_0}{T} = 1$ in isothermal reactors. Hence, we only need to include the T_0/T term when modeling a non-isothermal reactor.

in other words: in the absence of pressure drop

$$C_j = \frac{C_{LR,0}\left(\Theta_j - \frac{\nu_j}{\nu_{LR}} X\right)}{1 + \varepsilon X}\left(\frac{T_0}{T}\right)$$

if the reactor is additionally isothermal,

$$C_j = \frac{C_{LR,0}\left(\Theta_j - \frac{\nu_j}{\nu_{LR}} X\right)}{1 + \varepsilon X}$$